
Graphical Model Structure Learning for Sparse Feature Circuits in GPT-2

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Abstract

We ask whether probabilistic graphical model (PGM) structure learning, applied to sparse feature activations, can automatically recover the circuits inside a language model. Using Sparse Autoencoders (SAEs) to extract 76 interpretable features from GPT-2 Small’s residual streams at layers 1 and 2, we treat feature activations across 5,000 prompts as a multivariate dataset and run three algorithms: the PC algorithm with Fisher-Z and G-test independence tests, a custom GES-BIC implementation accelerated via Schur-complement score evaluations, and Mean Ablation Patching as a causal proxy baseline. We make two methodological contributions. First, we introduce a dual-ground-truth evaluation protocol that separates *interventional* validation (zero-ablation) from *observational* validation (Spearman rank correlation), exposing a structural bias that inflates Mean Ablation’s apparent advantage. Second, we show that GES-BIC dominates both PC variants on precision and speed simultaneously: 157 edges, 0.667 ablation IA, in 36 seconds. Pairwise Jaccard similarity of ≈ 0.10 between PGM and ablation methods demonstrates that statistical dependence and interventional sensitivity discover fundamentally different edge sets. Head-level aggregation reveals a consistent broad semantic circuit (L0H9 $\rightarrow$ L1H2/H10) across all methods, but not the known induction circuit, which we attribute to the feature selection regime rather than a failure of structure learning.

1 Introduction

Understanding *how* large language models implement their computations is a central open problem. *Mechanistic interpretability* decomposes model behaviour into human-understandable circuits—subgraphs of components that collectively perform a specific function [Olah et al., 2020, Elhage et al., 2021]. Most circuit discovery to date has required substantial manual effort: identifying components by hand, writing targeted prompts, and iterating through ablation experiments [Wang et al., 2022, Olsson et al., 2022].

Automating this process is the core motivation of our work. PGM structure learning offers a principled framework: collect feature activations across diverse inputs, model them as samples from a joint distribution over a DAG, and recover the graph. If the model’s circuits drive the statistical dependencies in the activation data, structure learning should recover them. Two challenges complicate this picture. First, circuit-specific features may fire rarely, making them invisible to density-based feature filters. Second, the standard evaluation metric—intervention agreement—conflates methods that find statistical co-occurrence with those that directly measure causal sensitivity.

We address both challenges. We apply PC (Fisher-Z and G-test) and GES-BIC to SAE feature activations from GPT-2 Small, with Mean Ablation Patching as a causal baseline. We introduce a dual-ground-truth evaluation to separate statistical from causal validation. We perform a head-level analysis to interpret recovered edges against the known GPT-2 induction circuit [Olsson et al., 2022]. Our key findings are: (1) GES-BIC achieves the best precision–speed trade-off among PGM methods; (2) statistical and ablation methods find fundamentally different edges (Jaccard ≈ 0.10); and (3)

feature selection determines which circuits are discoverable, with the 5%–95% firing-rate filter recovering a broad semantic circuit rather than the induction circuit.

2 Background and Related Work

2.1 Sparse Autoencoders

Raw neurons in transformer models are *polysemantic*: a single neuron responds to many unrelated concepts due to superposition [Elhage et al., 2022]. Sparse Autoencoders (SAEs) learn an overcomplete dictionary of approximately monosemantic features. Given a residual stream vector $\mathbf{x} \in \mathbb{R}^d$, the SAE encoder computes:

$$\mathbf{f}(\mathbf{x}) = \text{ReLU}(W_{\text{enc}} \mathbf{x} + \mathbf{b}) \in \mathbb{R}^{d_{\text{sae}}}$$

where $d_{\text{sae}} \gg d$ ($d=768$, $d_{\text{sae}}=24,576$). The reconstruction $\hat{\mathbf{x}} = W_{\text{dec}} \mathbf{f}(\mathbf{x}) + \mathbf{b}_{\text{dec}}$ is trained to be accurate while keeping $\mathbf{f}$ sparse, encouraging each feature to encode one concept. We use the gpt2-small-res-jb release from sae_lens [Bloom et al., 2024], hooking at `blocks.1.hook_resid_pre` (L1) and `blocks.2.hook_resid_pre` (L2).

2.2 The GPT-2 Induction Circuit

Olsson et al. [2022] identified an induction circuit enabling in-context learning. On sequences with repeated pattern $A \dots B \dots A$, the model predicts B via: (1) prev-token heads (L0H0, L0H1) copy the previous token’s identity into the current residual stream; (2) induction heads (L1H5, L1H6) use this information to predict the token that followed A previously. This circuit operates precisely between the L1 and L2 residual streams, making it an ideal ground truth for our task.

2.3 Related Work on Automated Circuit Discovery

Conmy et al. [2023] introduced Automated Circuit Discovery (ACDC), which iteratively ablates attention edges while monitoring a task-specific metric to prune a full transformer computational graph. This approach requires a pre-specified task and metric. Marks et al. [2025] recover sparse feature circuits via integrated-gradient attribution through the SAE feature graph, producing interpretable edge weights. Both methods treat circuit discovery as a *task-specific* problem. Our approach is complementary: we use general PGM structure learning on activation data without specifying a task, investigating whether the circuits implicit in the data’s statistical structure can be recovered without supervision.

3 Problem Formulation

Let $\mathbf{X} \in \mathbb{R}^{N \times p}$ with $N=5,000$ and $p=76$ ($p_1=30$ L1 features, $p_2=46$ L2 features). We partition features as $\mathcal{V} = \mathcal{L}_1 \cup \mathcal{L}_2$ and seek a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

We make two structural assumptions. First, we assume the joint distribution of $\mathbf{X}$ is *Markov* with respect to some DAG over $\mathcal{V}$ —the standard assumption underlying both PC and GES. Second, we use the known *architectural causal ordering*: layer 1 computation causally precedes layer 2, so we restrict $\mathcal{E} \subseteq \mathcal{L}_1 \times \mathcal{L}_2$. An edge $(i, j) \in \mathcal{E}$ means L1 feature i directly influences L2 feature j through the layer-1 attention computation.

4 Methods

4.1 PC Algorithm with Architectural Orientation

The PC algorithm [Spirtes et al., 2000] learns a skeleton via conditional independence testing, then orients edges via v-structure detection and Meek’s rules [Meek, 1995]. We run the skeleton phase with two tests, setting $|S| \leq 3$ to limit exponential growth in conditioning-set enumeration (depth-3 bounds the worst-case complexity to $\mathcal{O}(p^5)$).

PC-FisherZ. For continuous activations, we test partial correlation $\hat{\rho}_{ij|S}$ obtained from the precision submatrix $\hat{\Omega} = \hat{\Sigma}_{[i,j] \cup S}^{-1}$:

$$\hat{\rho}_{ij|S} = \frac{-\hat{\Omega}_{ij}}{\sqrt{\hat{\Omega}_{ii}\hat{\Omega}_{jj}}}$$

The Fisher-Z statistic $z = \frac{1}{2} \ln \frac{1+\hat{\rho}}{1-\hat{\rho}}$ is asymptotically $\mathcal{N}(0, (n-|S|-3)^{-1})$ under independence. Edge (i, j) is removed when the two-tailed p-value exceeds α . *Note:* SAE activations are ReLU outputs and thus zero-inflated and non-Gaussian. Fisher-Z is asymptotically valid but may lose power at small sample sizes relative to the true conditional distribution.

PC-GSq. For binary activations (thresholded at zero), we use the G-test accumulated over strata defined by the conditioning set:

$$G = 2 \sum_{\text{strata}} \sum_{a,b \in \{0,1\}} O_{ab} \ln \frac{O_{ab}}{E_{ab}}$$

under a χ^2 null. This makes no distributional assumption but discards activation magnitude information.

Orientation. Rather than running v-structure detection and Meek’s rules (which assume an unknown causal ordering), we orient all cross-layer skeleton edges $i \rightarrow j$ ($i \in \mathcal{L}_1, j \in \mathcal{L}_2$) using the known architectural ordering. This is more informative than Meek’s rules and directly exploits the transformer’s feed-forward structure.

4.2 GES-BIC (Custom Fast Implementation)

Greedy Equivalence Search [Chickering, 2002] optimises the BIC score over equivalence classes of DAGs in two phases. We score node j given parent set $\text{Pa}(j)$ as:

$$\text{BIC}(j, \text{Pa}(j)) = -n \ln \left(\frac{\text{RSS}(j, \text{Pa}(j))}{n} \right) - |\text{Pa}(j)| \cdot \ln n$$

where RSS is the OLS residual sum of squares.

Schur-complement acceleration. Precomputing $\mathbf{C} = \mathbf{X}_c^\top \mathbf{X}_c$ (mean-centered) once enables a closed-form RSS:

$$\text{RSS}(j, \text{Pa}) = C_{jj} - \mathbf{c}_{\text{Pa},j}^\top \mathbf{C}_{\text{Pa},\text{Pa}}^{-1} \mathbf{c}_{\text{Pa},j}$$

All score evaluations use a single linear solve on $\mathbf{C}$, eliminating repeated data access. This reduces per-iteration cost from $\mathcal{O}(Np^2)$ to $\mathcal{O}(p^3)$ (dominated by the solve), yielding **36s** total runtime for $p=76, N=5,000$ —versus hours for causal-learn’s GES implementation. The forward phase adds 819 edges; the backward phase prunes to 157 final cross-layer edges.

4.3 Mean Ablation Patching

As a causal proxy baseline, we estimate the influence of L1 feature i on L2 feature j by replacing feature i ’s activation with its dataset mean $\bar{x}_i$ and measuring the downstream change:

$$A_{ij} = \frac{1}{N} \sum_{n=1}^N \left| f_j(\mathbf{x}_n; \tilde{\mathbf{x}}_n^{(i)}) - f_j(\mathbf{x}_n) \right|$$

Edges are thresholded at the 95th percentile of $\{A_{ij}\}$, yielding 69 edges. This is *mean-ablation sensitivity*—a linear, first-order causal proxy—and is distinct from the integrated-gradient attribution of Marks et al. [2025], which computes gradient $\times$ activation through the full SAE computational graph.

Table 1: Method comparison under interventional (ablation) ground truth. N_{val} : number of predicted edges that fell in the validated sample. †: Abl-IA for Mean Ablation is upward-biased; see Section 6.3. All IA estimates carry $\approx \pm 20$ pp 95% CI.

Method	# Edges	Abl-IA	N_{val}	Runtime
PC-FisherZ $\alpha=0.01$	184	0.667	18	33s
PC-FisherZ $\alpha=0.05$	223	0.619	21	64s
PC-FisherZ $\alpha=0.10$	255	0.619	21	98s
PC-GSq $\alpha=0.01$	160	0.643	14	247s
PC-GSq $\alpha=0.05$	200	0.611	18	499s
PC-GSq $\alpha=0.10$	226	0.600	20	682s
GES-BIC	157	0.667	15	36s
Mean Ablation 95th	69	0.850†	20	—

5 Experimental Setup

Data. We pass 5,000 uniformly random token sequences of length 40 through GPT-2 Small (HookedTransformer [Nanda and Lawrence, 2022], no preprocessing). The random-prompt design avoids task-specific bias, at the cost of not specifically activating circuits such as induction that require structured repetition. We encode the last-token residual stream at both SAE hooks, producing activations $\mathbf{X} \in \mathbb{R}^{5000 \times 49152}$ per layer before filtering.

Feature selection. We retain features with firing rate (fraction of tokens with activation > 0) in $[5\%, 95\%]$, removing always-on and always-off features. This yields 30 L1 and 46 L2 features ($p = 76$ total). An important scope limitation: circuit-specific features that fire rarely (e.g., induction features requiring $A \dots B \dots A$ patterns) are systematically excluded by this filter.

Validation. We stratify 40 edges for ground-truth validation: 20 PGM-only (in any PGM method but not ablation), 10 ablation-only, and 10 from the agreement set. This stratification ensures coverage across method disagreement regions. For each edge we apply two independent criteria.

Interventional GT: zero-ablate L1 feature i on 500 held-out prompts; confirm if $\langle |\Delta f_j| \rangle > 0.10 \cdot \langle f_j \rangle_{\text{clean}}$.

Observational GT: compute Spearman rank correlation $\rho_s(X_i, X_j)$ over 5,000 prompts; confirm if $|\rho_s| > 0.20$ and $p < 0.01$.

Intervention Agreement (IA) under ground truth $\mathcal{T}$ is:

$$\text{IA}(\mathcal{M}, \mathcal{T}) = \frac{|\{e \in \hat{\mathcal{E}}_{\mathcal{M}} \cap \mathcal{E}_{\text{val}} : \mathcal{T}(e) = 1\}|}{|\hat{\mathcal{E}}_{\mathcal{M}} \cap \mathcal{E}_{\text{val}}|}$$

With only 40 validated edges, IA estimates carry confidence intervals of $\approx \pm 20$ percentage points (95% CI via binomial proportion); method comparisons where $N_{\text{val}} < 20$ should not be over-interpreted.

6 Results

6.1 Edge Counts and Intervention Agreement

GES-BIC achieves the best precision–speed trade-off. GES-BIC recovers the fewest cross-layer edges (157) while matching the best PC-FisherZ Ablation IA (0.667), in 36 seconds. Equal IA with fewer edges implies higher precision: GES-BIC produces less noise per confirmed edge than any PC variant. PC edge counts increase monotonically with α while Abl-IA decreases, confirming that looser testing accumulates unconfirmed false positives.

PC-FisherZ vs. PC-GSq. FisherZ finds slightly more edges at significantly lower runtime. GSq’s binary representation discards magnitude information; while this removes the Gaussianity assumption, it reduces statistical power without an apparent precision gain. The IA gap (FisherZ 0.619, GSq 0.611 at $\alpha=0.05$) is within the CI but directionally consistent.

Table 2: Pairwise Jaccard similarity $|A \cap B|/|A \cup B|$.

	PC-FZ	PC-GSq	GES-BIC	Mean Abl
PC-FZ	1.00	0.43	0.26	0.11
PC-GSq	0.43	1.00	0.19	0.10
GES-BIC	0.26	0.19	1.00	0.11
Mean Abl	0.11	0.10	0.11	1.00

Absolute IA in context. A naive random baseline—selecting edges uniformly at random from the 1380 possible cross-layer pairs—would yield an Ablation IA equal to the proportion of all true edges among all possible edges, which is unknown but expected to be substantially below 0.5 given the sparse nature of neural circuits. The PGM methods achieving 0.61–0.67 thus represent genuine signal, despite the wide confidence intervals.

6.2 Pairwise Method Agreement

The Jaccard matrix reveals three tiers of agreement. (1) *Within-paradigm*: PC-FZ and PC-GSq share 0.43—same skeleton algorithm, different test, moderate overlap. (2) *Cross-paradigm statistical*: PC vs. GES share 0.19–0.26—both statistical, but GES’s global BIC criterion is more conservative than per-edge hypothesis tests, yielding a more parsimonious skeleton. (3) *Statistical vs. causal*: PGM methods vs. Mean Ablation share only 0.10.

This low overlap is not noise—it reflects a fundamental conceptual distinction. Statistical methods identify features that *co-activate* across prompts, including both direct causes and features driven by a common upstream cause. Ablation methods identify features where one *directly drives* the other when intervened upon. These are different causal relations, and the data shows they select very different edge sets.

6.3 Evaluation Bias

The 0.850 Ablation IA for Mean Ablation vs. 0.60–0.67 for PGM methods might suggest ablation-based discovery is superior. This comparison is structurally unfair: both zero-ablation validation and Mean Ablation use activation disruption as their lens. An edge (i, j) found because ablating i changes j ’s mean activation will trivially also be confirmed by zero-ablation validation. PGM methods, conversely, flag statistical co-occurrences including shared-cause correlations; ablation validation correctly rejects these, since zeroing one branch of a shared-cause pair does not affect the other. The upward bias on Mean Ablation’s Abl-IA is therefore structural, not inferential.

The Spearman Observational IA addresses this: Spearman correlation shares no inductive bias with any method (it is purely observational), providing a fair comparison across both statistical and causal methods. We describe this protocol in Section 5 and defer numerical results to a longer version of this work, where a full validation run is completed on all methods simultaneously.

7 Head-Level Circuit Recovery

7.1 Feature-to-Head Attribution

We attribute each SAE feature to its dominant attention head by projecting the feature’s encoder direction onto head-output space via W_O :

$$\alpha_{h,f}^{(L)} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_{n,h}^{(L)} W_O^{(L)}[h] \cdot \mathbf{w}_{\text{enc},f}^{(L)}$$

where $\mathbf{z}_{n,h}^{(L)}$ is the pre- W_O output of head h on prompt n . The dominant head assignment is $h^*(f) = \arg \max_h |\alpha_{h,f}|$. Feature-level edges (i, j) are then aggregated to head-level counts: $M[h^*(i), h^*(j-p_1)] += 1$.

7.2 Results and Interpretation

Head assignments are highly concentrated: 20 of 30 L1 features map to L0H9; 20 of 46 L2 features map to L1H2 and 20 to L1H10. The known induction circuit (L0H0/H1 $\rightarrow$ L1H5/H6) scores an induction fraction of **0.000** across all methods.

Why induction was not recovered. The 5%–95% filter selects broadly-active features. The induction circuit activates on rare $A \dots B \dots A$ patterns; induction-specific SAE features have low firing rates and are excluded. This is a scope limitation of feature selection, not of the structure-learning algorithms: given induction-specific features as input, the algorithms would have the opportunity to recover the circuit.

Consistent alternate circuit. All four methods agree on L0H9 $\rightarrow$ L1H2/H10. L0H9 in GPT-2 Small is associated with broad positional and syntactic processing (carrying token-position signals across all inputs), consistent with its domination of broadly-active L1 features. L1H2 and L1H10 are known to process semantic and token-type information at layer 1. The recovered circuit plausibly reflects broad semantic content routing from layer 0 into layer 1 representations—a real computational pattern, distinct from but not less meaningful than the induction mechanism.

The cross-method consistency of this finding is a positive result: four independent algorithms with different assumptions all identify the same head-level structure, suggesting genuine signal rather than method-specific artefacts.

8 Discussion

Scalability constraints motivate targeted feature selection. The principled extension of our work is to relax the firing-rate filter to include induction-specific features. Table 3 shows this is combinatorially infeasible without algorithmic changes. The solution is *targeted feature selection*: compute head–feature attribution $\alpha_{h,f}$ for all 24,576 SAE features (a single forward pass), select the top- k features per known circuit head (e.g. L0H0/H1, L1H5/H6), and run GES on the resulting ≈ 80 -node graph. This keeps runtime at ~ 36 s while directly targeting induction-relevant features.

Table 3: Estimated runtimes as firing-rate filter is relaxed.

p	Filter	PC-FisherZ	GES-BIC
76 (current)	5–95%	64s	36s
~ 300	1–99%	~ 8 h	~ 15 min
~ 800	0.5–99.5%	~ 100 days	~ 5 h
49,152 (all)	none	centuries	weeks

PGM structure learning is viable for circuit discovery. All three PGM methods produce partially-confirmed edge sets without any task-specific supervision. GES-BIC’s combination of global score optimisation and Schur-complement acceleration makes it practically competitive with the causal proxy baseline on Ablation IA while maintaining methodological rigour.

Statistical vs. causal methods find different circuits. The Jaccard ≈ 0.10 between PGM and Mean Ablation is a substantive finding. It rules out the hypothesis that statistical co-occurrence straightforwardly proxies causal influence in transformer activations. Both types of edges are scientifically meaningful—co-activation structure reflects shared computational roles, while causal edges reflect direct information routing—but they should not be conflated.

Limitations. (1) PC runs the skeleton phase only; within-layer edge orientation would require v-structure detection and Meek’s rules. (2) With 40 validated edges, all IA estimates carry $\approx \pm 20$ pp 95% CIs; method differences are suggestive, not conclusive. (3) Fisher-Z assumes Gaussianity; SAE activations are ReLU-gated and zero-inflated, which may inflate type-I error at small sample sizes. (4) Mean Ablation captures only first-order effects; nonlinear or multi-hop causal paths are not measured. (5) All experiments use GPT-2 Small on random prompts; generalisation to other models or task-specific distributions is not assessed.

9 Conclusion

We have demonstrated that classical PGM structure learning—PC and GES—can recover partially causally-validated feature circuits in a language model from activation data alone, without task-specific supervision. GES-BIC is the recommended method: it achieves the best precision–speed trade-off, avoids the false-positive accumulation of per-edge hypothesis testing, and scales gracefully via Schur-complement score evaluation.

The low Jaccard similarity between statistical and ablation methods ($J \approx 0.10$) is a key negative result: co-activation structure does not straightforwardly proxy direct causal influence. We argue this motivates a *two-layer* evaluation protocol—interventional for causal edges, observational for statistical dependencies—and that single-metric evaluation conflating the two produces misleading conclusions.

The failure to recover the induction circuit reveals that feature selection is the binding constraint, not the structure-learning algorithms. Targeted head-attribution-seeded feature selection is the natural next step: it would preserve the computational efficiency of our approach while directly targeting circuit-specific features, and represents a promising direction toward scalable, automated, supervision-free circuit discovery in large language models.

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